



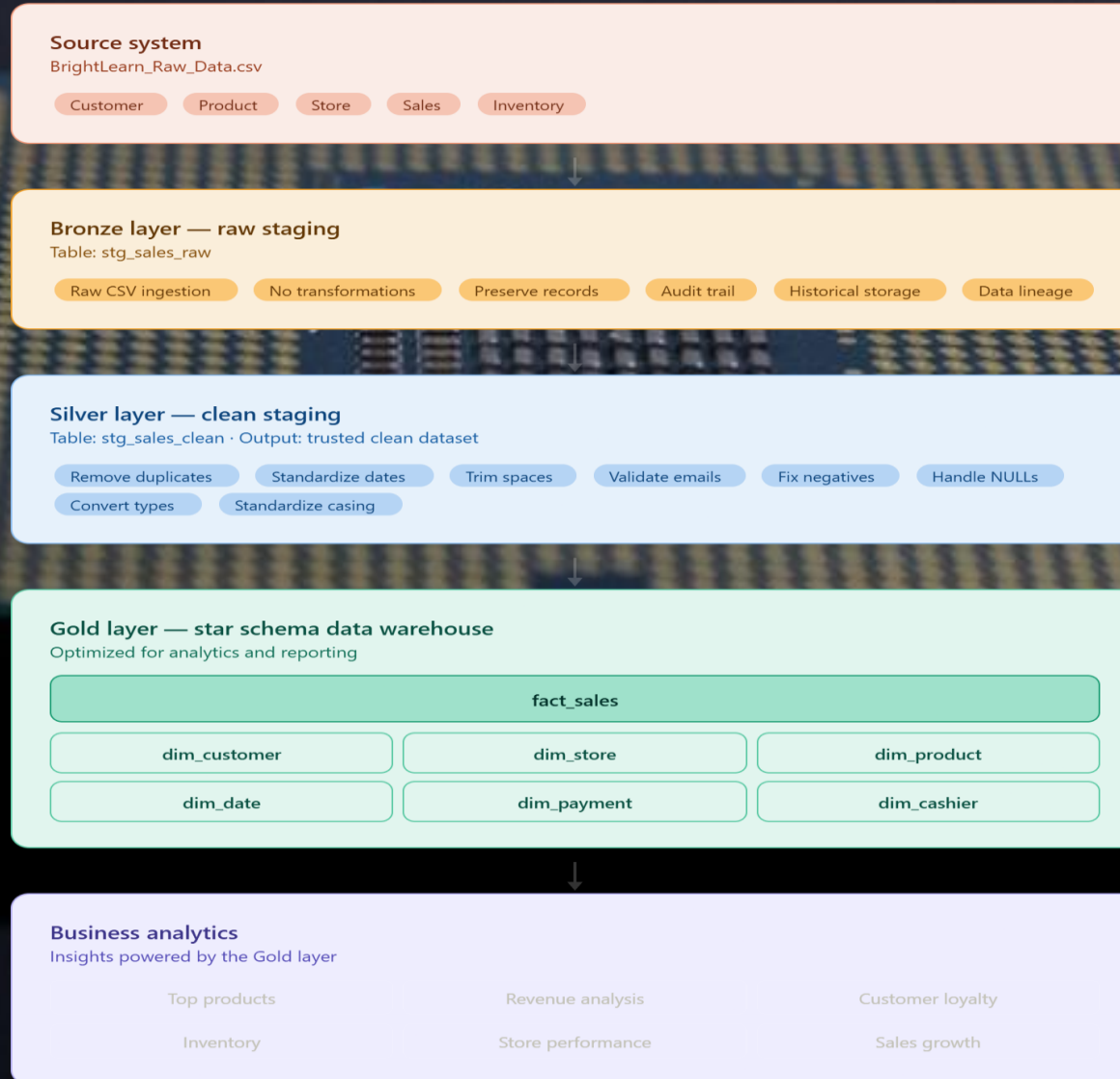
BrightLearn Data Warehouse Build

Building a Modern Data
Warehouse Using
Medallion Architecture,
SSIS, and SQL Server

Business Problem

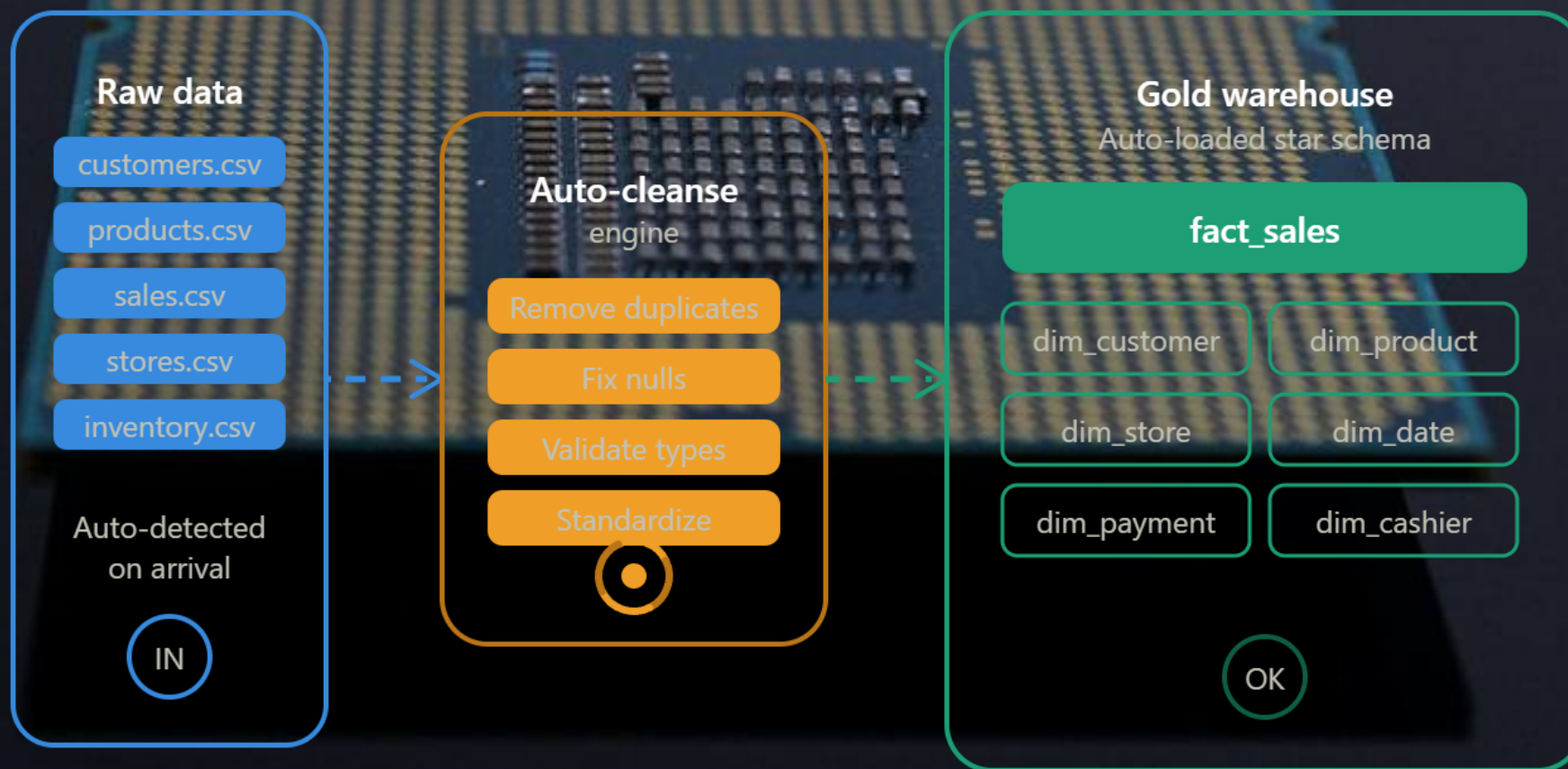
Single flat POS file, poor data quality, no reporting capability.

Medallion Architecture



SSIS ETL Pipeline

Automated ingestion, cleansing and warehouse loading.



Data Quality Findings

BrightLearn — real data issues found & resolved

5 000 rows · 29 columns · actual records shown below

Duplicate transactions

Same customer · same date · same total

2024-02-11 Sipho Zulu R502.92 Tomato Sauce

2024-02-11 Sipho Zulu R502.92 White Bread

2024-02-11 Sipho Zulu R502.92 Rooibos Tea

2024-02-11 Sipho Zulu R502.92 Instant Coffee

These are basket line items, not duplicate rows.
transaction_id missing — dedupe by full key.

Resolved: group by transaction_date + customer + amount ✓

Null customer fields

253 rows — name, email, phone, city all NULL

Row 94: NULL NULL NULL NULL NULL

Row 95: NULL NULL NULL NULL NULL

Also: 811 rows missing customer_email alone

e.g. Musa Ntuli · musa.ntuli@gmail.com → NaN

Full-null rows → flag as walk-in customers.
Email-only nulls → fill "unknown@brightlearn.co.za".

Resolved: default values + walk-in customer flag ✓

Mixed date formats

Same column — 3 formats in the same file

2024-06-12 00:00:00

2024-06-12

17/05/2024

2024-05-17

13-04-2024

2024-04-13

pd.to_datetime(errors='coerce') parses all formats.
Output: YYYY-MM-DD ISO 8601 standard.

Resolved: all dates → 2024-MM-DD format ✓

Negative amounts & null categories

Bad values corrupting totals

Annah Molefe 17/05/2024 -R40.02

Kavitha 2024-05-11 -R201.00

Busisiwe Nene — category: NULL

132 rows with NULL category field

Negative amounts → flag as refunds (not errors).
Null category → map from SKU prefix (GRO / ELE).

Resolved: refund flag + SKU-based category fill ✓

● Duplicates

● 253 null rows

● Mixed dates

● Bad values

Business Value

8 BUSINESS QUESTIONS FROM THE SCOPE

Turning Data into Insights. Driving Better Business Decisions for BrightLearn.

01 Top 5 Best-Selling Products by Revenue

- OBJECTIVE**
Identify the products generating the highest revenue.
- KPI**
Total Revenue by Product
- BUSINESS INSIGHT**
Highlights the five products contributing the most revenue between January and June 2024.
- POTENTIAL ACTIONS**
- Increase stock levels for high-demand products
 - Create product bundles
 - Launch promotional campaigns

02 Total Revenue per Store, Broken Down by Month

- OBJECTIVE**
Measure revenue generated by each store on a monthly basis.
- KPI**
Monthly Revenue per Store
- BUSINESS INSIGHT**
Provides a month-by-month view of revenue generated by each branch between January and June 2024.
- POTENTIAL ACTIONS**
- Reward high-performing stores
 - Investigate underperforming locations
 - Improve regional sales strategies

03 Month-over-Month Revenue Growth Rate

- OBJECTIVE**
Measure sales growth trends over time.
- KPI**
Revenue Growth %
- FORMULA**
$$\frac{\text{Current Revenue} - \text{Previous Revenue}}{\text{Previous Revenue}} \times 100$$
- BUSINESS INSIGHT**
Identifies periods of growth or decline.
- POTENTIAL ACTIONS**
- Investigate revenue declines
 - Replicate successful sales campaigns
 - Improve forecasting accuracy

04 Top 10 Loyalty Customers by Total Spend

- OBJECTIVE**
Identify the most valuable loyalty customers.
- KPI**
Total Customer Spend
- BUSINESS INSIGHT**
Ranks loyalty customers according to total spending.
- POTENTIAL ACTIONS**
- Provide VIP rewards and benefits
 - Offer exclusive promotions
 - Run personalized marketing campaigns

05 Inactive Loyalty Customers (Win-Back Campaign)

- OBJECTIVE**
Identify loyalty customers who have not made a purchase since 28 April 2024.
- KPI**
Days Since Last Purchase
- BUSINESS INSIGHT**
Highlights loyalty members at risk of churn.
- POTENTIAL ACTIONS**
- Email campaigns
 - Discount vouchers
 - Loyalty point incentives
 - Personalized offers
- STRATEGIC VALUE:** Supports customer retention initiatives and reduces customer churn.

06 Average Transaction Value by Loyalty Tier

- OBJECTIVE**
Compare spending behavior across loyalty tiers.
- KPI**
Average Transaction Value (ATV)
- FORMULA**
$$\frac{\text{Total Revenue}}{\text{Number of Transactions}}$$
- BUSINESS INSIGHT**
Determines whether Gold, Silver, or Bronze customers generate the highest transaction values.
- POTENTIAL ACTIONS**
- Redesign loyalty benefits
 - Encourage upgrades to higher tiers
 - Target specific customer segments

07 Total Quantity Sold per Product Category, per Store

- OBJECTIVE**
Analyze product demand across stores and categories.
- KPI**
Quantity Sold
- BUSINESS INSIGHT**
Measures how many units were sold for each product category in every store.
- POTENTIAL ACTIONS**
- Improve inventory distribution
 - Optimize store layouts
 - Customize regional product offerings

08 Inventory Reorder Analysis

- OBJECTIVE**
Identify products requiring replenishment.
- KPI**
Current Stock vs Reorder Threshold
- BUSINESS INSIGHT**
Identifies products whose inventory levels have fallen below predefined reorder thresholds.
- POTENTIAL ACTIONS**
- Generate purchase orders
 - Replenish inventory
 - Prevent stockouts
 - Improve supply chain planning



EXECUTIVE SUMMARY

These 8 business questions enable BrightLearn to make data-driven decisions that improve sales performance, customer loyalty, and operational efficiency.



SALES INTELLIGENCE

Top products identified, revenue trends monitored, store performance measured.



CUSTOMER INTELLIGENCE

High-value customers identified, customer retention opportunities uncovered, loyalty program effectiveness evaluated.



INVENTORY INTELLIGENCE

Stock shortages detected, reorder requirements highlighted, inventory optimization enabled.



STRATEGIC BENEFITS

Improved decision-making, better customer retention, increased revenue visibility and operational efficiency.

Technology Stack



```
SELECT c.Name,  
       e.Department,  
       COUNT(*) AS TotalEmployees  
FROM Employees e  
WHERE e.IsActive = 1  
GROUP BY c.Name, e.Department  
ORDER BY TotalEmployees DESC;
```

T-SQL

Conclusion

CONCLUSION



The proposed data warehouse for **BrightLearn** transforms raw operational data into a structured, analytics-ready environment that supports efficient business reporting and decision-making. By implementing a **star schema** consisting of centralized fact tables and well-defined dimension tables, the warehouse enables fast querying, simplified data analysis, and scalable reporting across **sales, customers, products, stores, payments, and inventory**.

KEY BENEFITS



Fast querying
Optimized star schema for high performance



Simplified analysis
Business-friendly structure for easy reporting



Scalable reporting
Supports growth across multiple business areas



Consistent & accurate
Clean, integrated and trusted data



Better decisions
Actionable insights across the business

BRIGHTLEARN DATA WAREHOUSE – KEY DATA DOMAINS

CUSTOMERS



- Customer profiles
- Contact details
- Location
- Loyalty tier
- Customer since

Understand customer segments and behavior

SALES



- Transactions
- Quantity sold
- Prices & discounts
- Net sales
- Profit

Track sales performance and profitability

PRODUCTS



- Product details
- Categories
- Sub-categories
- Supplier
- Cost & price

Analyze product performance and profitability

STORES



- Store details
- Location
- Region
- Manager
- Province

Compare store and regional performance

PAYMENTS



- Payment methods
- Cash
- Credit Card
- Debit Card

Analyze payment patterns and trends

INVENTORY



- Stock on hand
- Reorder threshold
- Product
- Store
- Date snapshot

Monitor inventory levels and optimize stock



This data warehouse provides a reliable, integrated, and analytics-ready foundation that turns raw data into meaningful insights for data-driven business decisions.